

Hospital Readmission Risk

1. Project Goal

The goal of this project is to analyze hospital patient data to identify key factors that contribute to 30-day readmissions. By examining patient demographics, clinical conditions, treatment details, and discharge outcomes, the project aims to uncover high-risk patient groups and provide insights that can help healthcare providers reduce avoidable readmissions and improve patient care.

2. Dataset Summary

The dataset contains patient-level hospital records used to study readmission patterns. It includes demographic information, clinical characteristics, hospital stay details, and discharge outcomes for each patient.

Key fields include:

- **Patient Information:** age, gender, region
- **Clinical Data:** primary diagnosis, comorbidities count
- **Hospital Stay Details:** length of stay, treatment type, medications count
- **Patient History:** previous readmissions, follow-up visits
- **Administrative Data:** insurance type, discharge disposition
- **Target Variable:** label (0 = no readmission, 1 = readmitted within 30 days)

3. Data Cleaning in Excel

Data cleaning and preparation were performed in Excel to ensure accuracy and consistency before analysis.

Steps performed:

- Validated categorical variables using pivot tables to identify inconsistencies (e.g., variations in season, diagnosis, and insurance labels)
- Checked for missing values and ensured key variables such as readmission label were complete
- Reviewed numerical ranges to confirm logical values (e.g., age, length of stay, comorbidities count)
- Created derived features to improve analysis:
 - Age groups
 - Length of stay buckets
 - Comorbidity categories
- Revalidated the dataset after cleaning to confirm consistency and accuracy

4. Data Analysis Using SQL

Overall Readmission Rate

```
SELECT  
  
ROUND(AVG(label) *100 ,2) AS readmission_rate  
  
FROM hospital_readmission  
  
;
```

readmission_rate
77.29

Readmission by Diagnosis

```

SELECT
primary_diagnosis,
COUNT(*) AS total_patient,
ROUND(AVG(label) *100 ,2) AS readmission_rate
FROM hospital_readmission
group by primary_diagnosis
order by readmission_rate desc

```

	primary_diagnosis	total_pati...	readmission_rate	
	Sepsis	497	86.92	
	COPD	952	86.87	
	Heart Failure	403	85.86	
	Stroke	883	85.28	
	Kidney Disease	868	83.29	
	Pneumonia	425	73.88	
	Appendicitis	534	71.16	
	Diabetes	1470	70.54	
	Fracture	434	70.28	
	Hypertension	1301	69.64	
	Influenza	233	68.67	

Readmission by Discharge Disposition

```
SELECT discharge_disposition,  
COUNT(*) AS total_patient,  
ROUND(AVG(label) *100 ,2) AS readmission_rate  
FROM hospital_readmission  
group by discharge_disposition  
order by readmission_rate desc  
;
```

	discharge_disposition	total_patient	readmission_rate
	Skilled Nursing	1568	92.98
	Home Health	6065	75.85
	Home	317	34.07
	Rehab	50	34.00

Readmission by Age Group

```
SELECT Age_group,  
COUNT(*) AS total_patient,  
ROUND(AVG(label) *100 ,2) AS readmission_rate  
FROM hospital_readmission  
group by Age_group  
order by readmission_rate desc  
;
```

	Age_group	total_patient	readmission_rate	
	80+	790	96.46	
	65-79	1922	94.22	
	50-64	2698	79.36	
	35-49	1875	61.76	
	18-34	715	43.50	

Readmission by Insurance Type

```

SELECT insurance_type,
COUNT(*) AS total_patient,
ROUND(AVG(label) *100,2) AS readmission_rate
FROM hospital_readmission
GROUP BY insurance_type
ORDER BY readmission_rate DESC;

```

	insurance_type	total_patient	readmission_rate	
	Medicare	2542	95.36	
	Uninsured	808	76.11	
	Medicaid	1598	70.59	
	Private	3052	66.06	

Length of Stay Impact

```

SELECT Stay_Bucket,
COUNT(*) AS total_patient,
ROUND(AVG(label) *100,2) AS readmission_rate

```

FROM hospital_readmission

GROUP BY Stay_Bucket

ORDER BY readmission_rate DESC;

	Stay_Bucket	total_patient	readmission_rate	
	8-14 days	4301	84.63	
	4-7 days	3682	68.96	
	15+ days	3	66.67	
	1-3 days	14	14.29	

Comorbidity Impact

SELECT comorbidities_count,

COUNT(*) AS total_patient,

ROUND(AVG(label) *100,2) AS readmission_rate

FROM hospital_readmission

GROUP BY comorbidities_count

ORDER BY readmission_rate DESC;

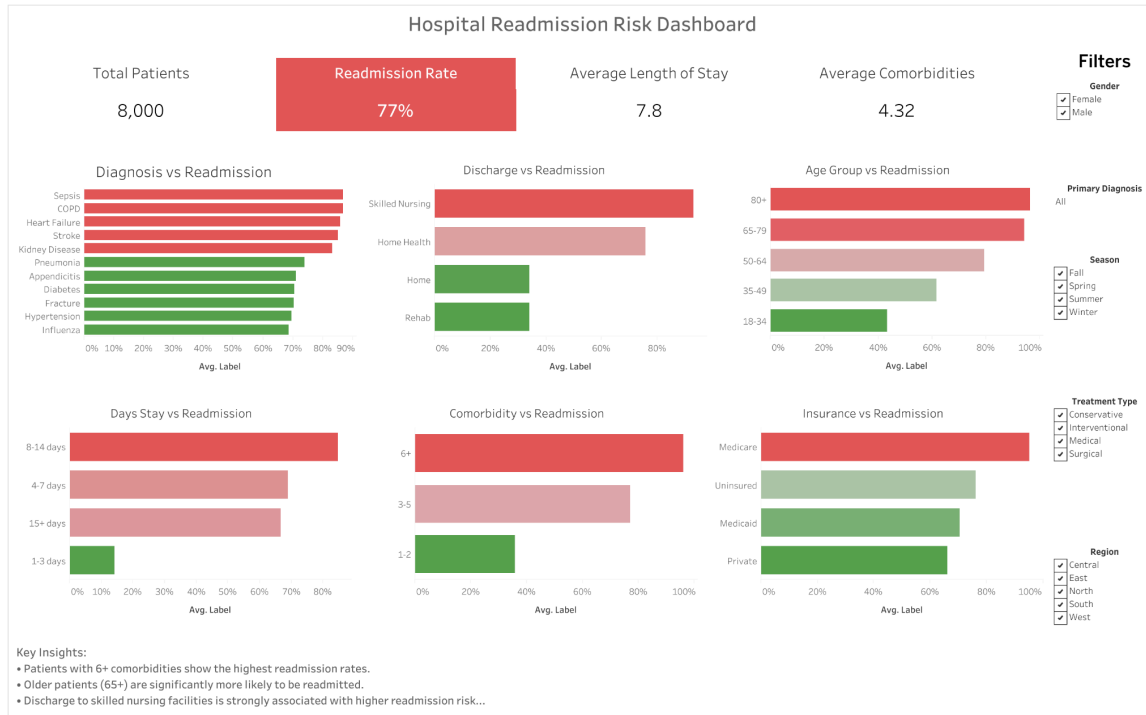
	comorbidities_count	total_patient	readmission_rate	
	10	2	100.00	
	8	70	97.14	
	7	371	96.23	
	6	1021	96.18	
	9	13	92.31	
	5	1960	91.79	
	4	2397	75.05	
	3	1520	61.32	
	2	534	38.95	
	1	112	21.43	

Previous Readmissions Effect

```
SELECT prev_readmissions,  
COUNT(*) AS total_patient,  
ROUND(AVG(label) *100,2) AS readmission_rate  
FROM hospital_readmission  
GROUP BY prev_readmissions  
ORDER BY readmission_rate DESC;
```

	prev_readmissions	total_patient	readmission_rate	
	5	24	100.00	
	4	320	96.25	
	3	622	95.34	
	2	2757	90.42	
	1	3800	69.53	
	0	477	25.79	

5. Dashboard in Tableau



6. Business Recommendations

- Implement targeted care plans for patients with 6+ comorbidities, as they show the highest readmission rates
- Provide closer monitoring and follow-up appointments after discharge
- Flag long-stay patients as high-risk before discharge
- Provide continuous care plans instead of one-time treatment
- Provide follow-up within 48–72 hours after discharge